PLATE 0 · TITLE

From generic chatbots to teacher-configurable AI tutors

Work in progress. The design and evaluation of Novedu in secondary education: alignment through teacher-owned configuration, curricular grounding, and in-region open-weight models.

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HTL Leonding, Austria · IMCL 2026

Generic chatbots in school: adopted, not aligned

94%

of our students have used an AI tool

84.5%

use one regularly

Survey, n = 284 students, HTL Leonding

OUR POSITION

Alignment by the school, not by model training

What Novedu adds to the classroom

Knows the lesson	topics, limits, and terms as taught in class
Guides, does not solve	questions and hints instead of finished answers
Owned by teachers	versioned, shared, reusable activity files
More than chat	curriculum-aware tutors, **new:** with a board for visualizations quizzes · writing coach · coding agents
School-grade defaults	exchangeable models · anonymous by default · aggregate oversight

Survey: three key requirements

#1

students' strongest requirement

Know my lesson

→ teacher-authored topics

85%

of teachers: most important property

No wrong answers

→ evaluations

6%

of teachers want to read chats; 76% want statistics

Oversight, not surveillance

→ aggregate statistics only

n = 318 (284 students, 34 teachers) at one technical secondary school; no direct generalization to general-education schools.

Agent-first: AI assistants write the activities

TEACHER BRINGS

Existing material

worksheets, PDFs, lesson notes

and / or

High-level teaching goals

“linked lists, Socratic, never show the solution”

→

GENERAL AI ASSISTANT

Writes the files

reads Novedu's documentation for agents, checks its work with Novedu's validator

→

NOVEDU

Activities

Fragment libraries

reusable behavior, shared across the school

Since submission. Non-technical teachers have adopted this workflow, too. There is room for improvement, but the direction works.

Demo

app.novedu.at

Open protocol, exchangeable models

Novedu
each activity names its model

→

OpenAI-compatible API
open protocol

→

Local
self-hosted, Austria

EU cloud
hyperscaler, European region

World
global providers

Small models work well

for Socratic teaching and coding beginners. No frontier model needed.

Evaluations make switching safe

Re-run the test scenarios of every tutor and quiz before changing provider or model, then switch with confidence.

Since submission: we did exactly that, moving from self-hosting to an EU cloud for stability.

Align AI tutors by configuration, not by training

RQ1 What do students and teachers require of an institutional AI tutor?

Four requirements: curricular grounding, factual reliability, configurable scaffolding, aggregated oversight

RQ2 How can a school align tutors through teacher-owned configuration rather than model training?

Treat alignment as a configuration problem: modular tutor files the teachers own, independent of the model provider

RQ3 Can a small open-weight model deliver Socratic tutoring, and how much prompt detail is needed?

Tentatively yes: on one topic, it scored at or above frontier baselines, if teachers supply detailed prompts

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PROJECT

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THANKS TO

the students who built Novedu, and SCCH for in-region inference

Q&A

Students already use generic chatbots. The open question is how a school channels them for school purposes.

94%

of our students have used an AI tool; **84.5%** regularly.

Pedagogically, a generic chatbot

- hands out answers instead of guiding the student
- knows neither the class's topic nor the teacher's intent

Institutionally, it

- sends student data to services the school does not control (GDPR, EU AI Act)
- gives teachers no oversight of its use in their subject

Our position. A school cannot realistically align tutors by *training* models. It can align them *institutionally*: teacher-owned configuration, curricular grounding, privacy defaults, and evaluation.

Why Novedu, and not just a generic chatbot?

EXAMPLE: A BIOLOGY TUTOR

> Student: “Why does the plant actually need light?”

NOVEDU ADDS THE TEACHER'S RULES

Content	photosynthesis, as covered in class
Limits	no formulas, no terms not yet taught
Feedback	hints instead of finished solutions

> Tutor: “Good question! What happens to a house plant that stands in a dark basement for weeks?”

What Novedu adds in the classroom	Mechanism and evidence
Knows the lesson	Students' top requirement (**M = 3.53**); teacher-authored topic modules
Guides, does not solve	Generic chatbots hand over answers [Sonkar 2024]; frontier models over-respond in our tests
Owned by teachers	Prompts as versioned, shared, reusable files, checked by an evaluation harness
More than chat	Graded open-answer quizzes, a writing coach, a proxy for coding agents
School-grade defaults	Exchangeable models, anonymous by default, aggregate oversight

To be fair: an IT expert can prompt a generic chatbot to do much of this. Most teachers cannot, and should not have to. Novedu does that engineering once, then shares, versions, and checks it for the whole school.

Three research questions, three contributions

	Question	Contribution
RQ1	What do students and teachers at a technical secondary school require of an institutional AI tutor?	Requirements evidence base (n = 318), traced to design decisions
RQ2	How can a school align tutors with those requirements through teacher-owned configuration rather than model training?	A teacher-configurable architecture over in-region, open-weight models
RQ3	Can a small, in-region open-weight model deliver Socratic tutoring, and how much prompt detail must teachers provide?	A reproducible evaluation harness with preliminary results

Survey: four requirements, each traced to a design

Survey signal	Design implication	Novedu response
Tutor must know my lesson content (**M = 3.53**)	Curricular grounding over generic chat	Teacher-authored topic and tutor modules
Wrong answers: students' top objection; teachers' top property (84.8%)	Reliability must be evaluated	Socratic tutor evaluation harness
“It depends, let me choose” the help style (52%)	One Socratic mode is too rigid	Reusable behavior fragments
Reject read-along (57%); teachers want aggregate stats (75.8%), not chats (6%)	Oversight must not be surveillance	Anonymous defaults, aggregate code statistics
Want training and templates; only 6% feel well-informed legally	Adoption depends on low effort	Prompt library, starter tutors, AI authoring aids

Survey. n = 318 (284 students, 34 teachers), supervised in-class pretest, 4-point Likert scale, no midpoint. Single site, 72.9% of students in the IT branch: no direct generalization to general-education schools.

A tutor is a file the teacher owns, and AI writes it

TEACHER'S EXISTING MATERIAL

worksheets, PDFs, lesson notes, converted by a general AI assistant

↓

TUTOR FILE, PER SUBJECT

Selects fragments, fills their inputs, adds final instructions, names the model

+

FRAGMENT LIBRARIES

Central, versioned modules: persona, “never give the solution”, topic limits

↓ assembled and validated at request time

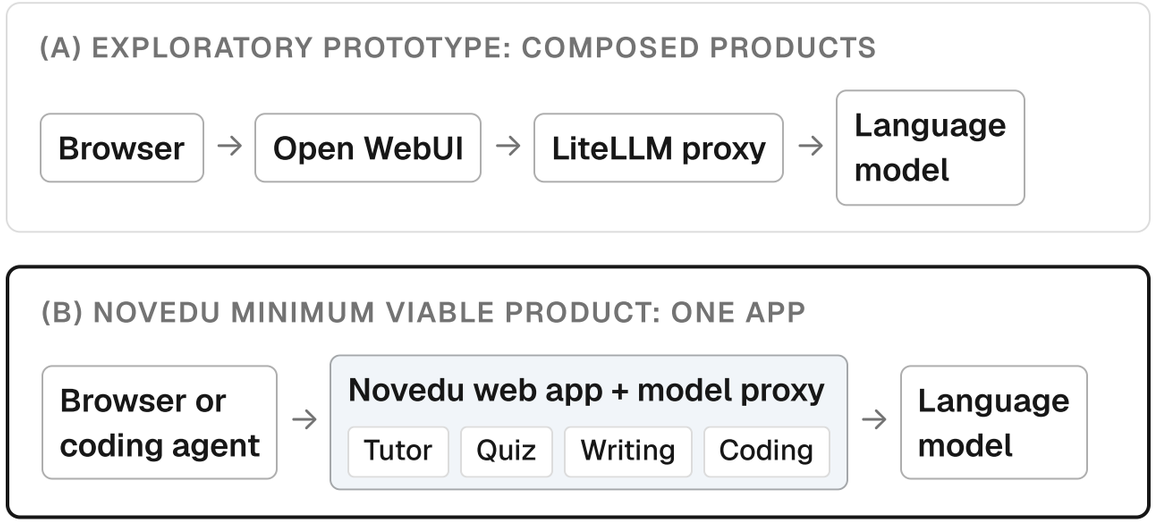
ONE SYSTEM PROMPT → ONE AGENT

served by the model the file names

```yaml
id: ts-sorting-algorithms
llm:
  model: RedHatAI/gemma-4-31B-it-FP8-Dynamic
prompt:
  fragment_files:
    - id: general_fragments
      url: "../shared/general-fragments.yaml"
  tutor_instructions: |
    {{fragment "general_fragments.socratic_tutor"}}
    {{fragment "general_fragments.topic_limits"
        allowed_topics=(array "Bubble Sort: …"
                              "Selection Sort: …")}}
```

SINCE SUBMISSION **Teachers no longer write these files by hand.** Even our non-technical language teachers turn their existing material into tutor files with a general AI assistant, and most never see the file itself. A graphical editor is no longer needed; Novedu validates and versions every save.

One abstraction, several activities

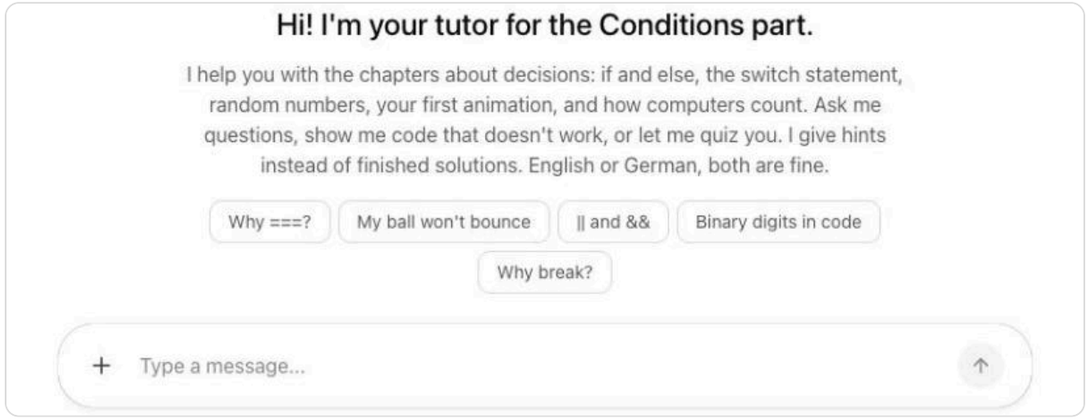


A teacher mints a short **code** that points at an activity file; one access-gating and statistics mechanism serves every kind.

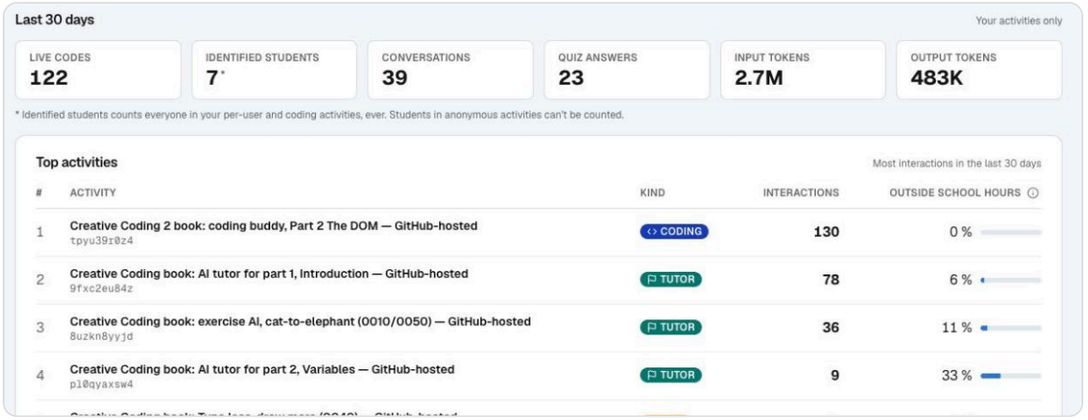
Activity	What the AI may do	Status
Tutor	Guide with questions and hints	in use
Quiz	Grade open answers against a server-only rubric	in use
Writing	Coach beside the draft; read, never edit	in use
Coding	Inject the teacher's rules into students' coding agents	implemented

Planned at submission	SINCE SUBMISSION
Smartphone support	implemented
Curriculum retrieval	implemented
Administration consoles	implemented
Graphical editor	dropped: AI assistants write the files

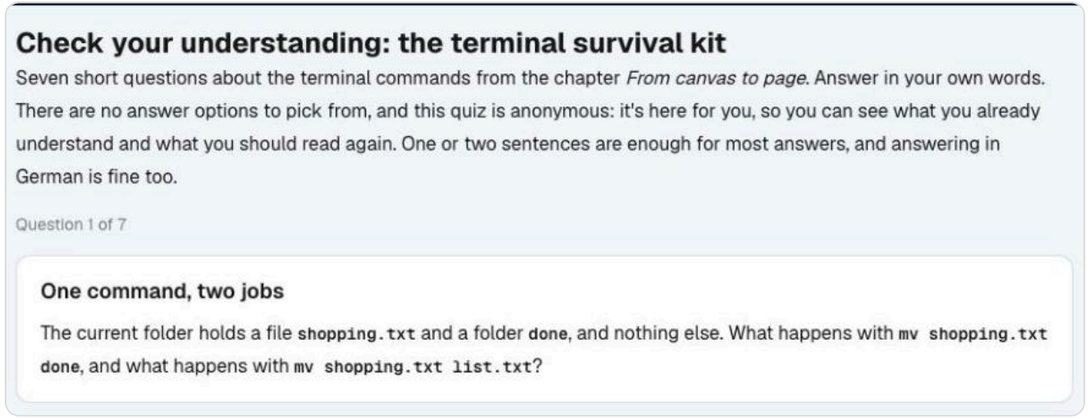
What students, teachers, and administrators see



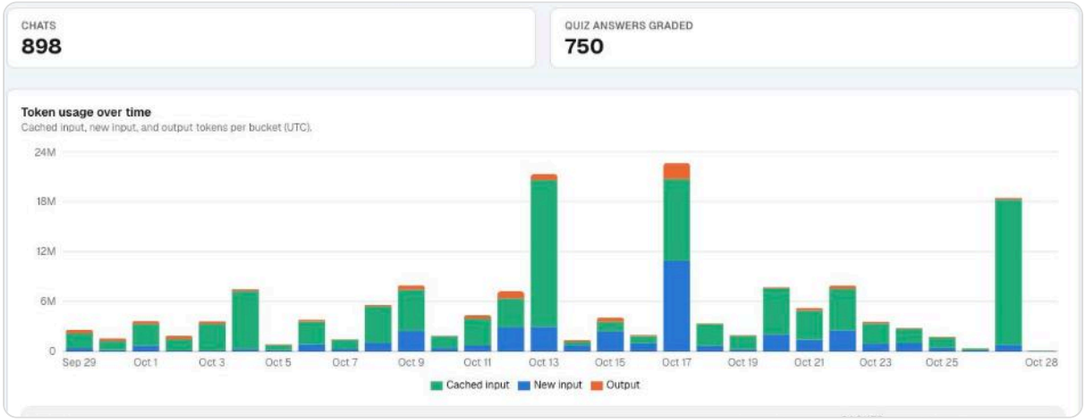
Student · Tutor: the teacher's activity and its scope



Teacher · Dashboard: aggregate statistics, never individual chats



Student · Quiz: open answers, graded against a hidden rubric



Administration · Usage: tokens per day across all activities

Deployment: the model provider is exchangeable

Inference, as submitted

Open-weight, quantized models, roughly 27–31 billion parameters

In-region GPU server of the Software Competence Center Hagenberg (SCCH), free within a government-funded project

One OpenAI-compatible interface, server-side only; each activity file names its model

Defaults

School single sign-on; teacher actions gated by group

Anonymous unless a teacher opts out; the quiz grader stores nothing

Telemetry carries no message content; chat history is not retained

SINCE SUBMISSION **Self-hosting had stability problems, so we moved temporarily to a hyperscaler's European region.** Because the model is configuration, the switch was a settings change, re-checked with the evaluation harness. Self-hosting remains our medium- to long-term strategy.

Privacy-aware design, not demonstrated legal compliance. Every inference provider, self-hosted or hyperscaler, is a separate processor. Enterprise assistants (e.g., Microsoft 365 Copilot) offer contractual guarantees of their own.

Evaluation: does the tutor behave as specified?

The teacher defines what is correct, in the tutor file and in a rubric per scenario. Within that frame, the evaluation measures the tutor's **behavior** in four steps:

1 · SCENARIO

A partial student–tutor conversation ending on a student turn

→

2 · REPLIES

Next tutor turn from **5 models × 3 prompt lengths** (6, 84, 172 lines)

→

3 · AI JUDGE

A language model (gpt-5.4, high effort) scores each reply against the rubric

→

4 · PASS

no negative criterion violated **and** ≥ 70% of positive criteria met

RUBRIC EXAMPLE

+ “asks one guiding question about the student's code”

– “states the corrected code”

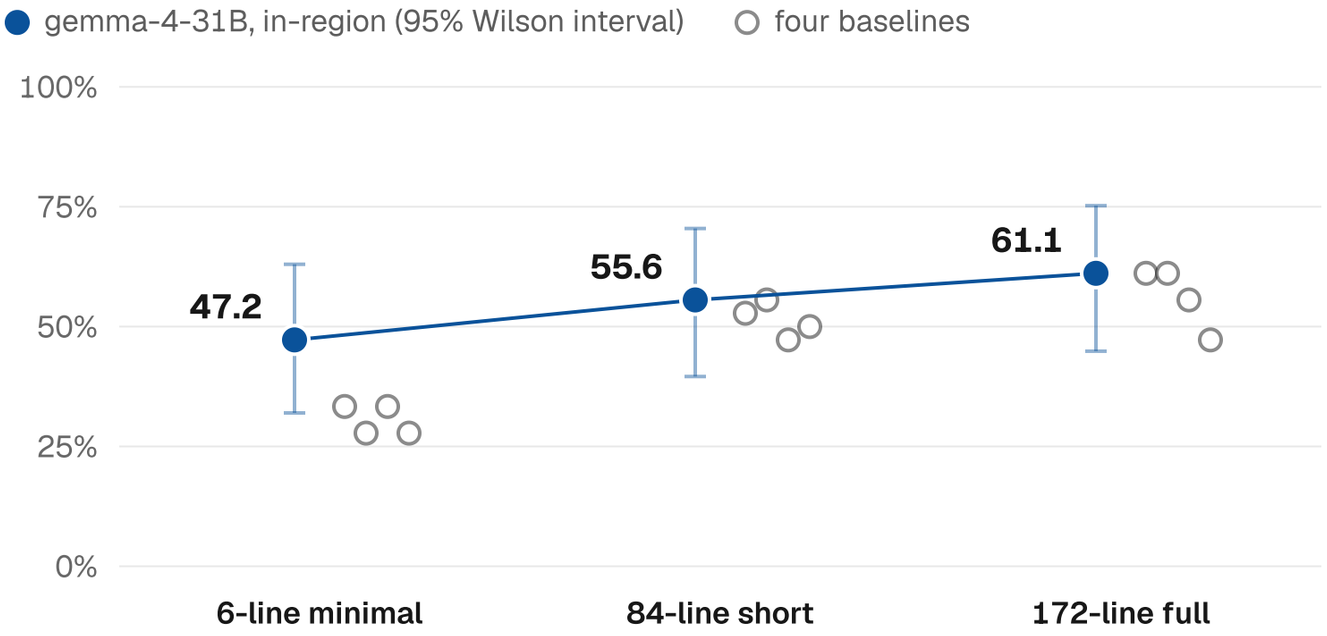
SOCRATIC VIOLATIONS, FIXED LIST

gave answer · multiple questions · lectured · judgmental · off-topic

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Design. One programming topic (linked lists), 12 scenarios × 3 replies = 36 judged replies per cell; each verdict quotes its evidence. The harness (SocraticTutorEval) re-runs automatically for any model or prompt variant. SINCE SUBMISSION it is how we re-checked our tutors when we changed provider.

Novedu works well with a small model



Prompt detail matters. The two-sentence prompt lowers the pass rate for every model: **−14** points in-region, up to −33 for gpt-5.4-mini.

A small model is enough here. gemma-4-31B matches the best frontier configurations on the full prompt (22/36) and is the most robust on the terse one (47.2% vs ≤ 33.3%).

Frontier models did not do better. Their failure mode: over-responding, handing over the full answer. Higher reasoning effort did not help.

Preliminary. n = 36 per cell; 95% Wilson intervals span about ±16 points, so differences within a prompt length are not statistically separable. One topic; an AI judge not yet validated against teachers. A descriptive observation, not evidence of equivalence.

As submitted, and what has changed since

As submitted	SINCE SUBMISSION
Not yet usable on smartphones	Smartphone support implemented
“Know my lesson” covered only by teacher prompts; retrieval planned	Curriculum retrieval implemented
Hand-written files are a barrier; graphical editor planned	AI assistants write the files; editor dropped
In-region inference at SCCH	A hyperscaler in Europe for now; self-hosting stays the goal
A small model was viable on one topic	Daily use agrees: smaller, cheaper models suffice
Harness: one topic, judge not yet validated	Next: validate the judge against teacher ratings, across subjects
Single-site survey; no learning-outcome data	Next: more schools; in-class comparison with a generic chatbot

A living system. The left column is the paper. The right column is practical experience since submission, not yet systematically evaluated; we will report on it at the next conference.

A school can align AI tutors by configuration, not by training

RQ1	What do students and teachers require of an institutional AI tutor?	Four requirements: curricular grounding, factual reliability, configurable scaffolding, aggregated oversight
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Table 2

Pass rate (%) and passes out of 36, by model and prompt length

Model	6-line	84-line	172-line
gemma-4-31B-FP8 (SCCH, in-region)	**47.2** (17)	55.6 (20)	**61.1** (22)
gpt-oss-120b (open weight, medium)	33.3 (12)	52.8 (19)	61.1 (22)
gpt-5.4-mini	27.8 (10)	55.6 (20)	61.1 (22)
gpt-5.4 (medium reasoning)	33.3 (12)	47.2 (17)	55.6 (20)
gpt-5.4 (high reasoning)	27.8 (10)	50.0 (18)	47.2 (17)

Linked-lists scenarios, each reply judged once by gpt-5.4 at high reasoning effort. With n = 36 per cell, 95% Wilson intervals span about ±16 points. Model IDs, decoding parameters, judge prompt, and tutor prompts are archived with the harness. The prompts predate the fragment model.

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